**Lab 4:** Applied Crypto & B.C
**Instructor:** Prof. Oumaima FADI
**Level:** Cybersecurity - 4th Year

# Lab 4: P-Box, S-Box and RSA Algorithms

## Objectives

- Understand the logic of RSA.
- Identify the main components and workflow of P-Box and S-Box.

## Instructions

- The lab report must be submitted one week after the lab session on the Moodle platform, respecting the deadline mentioned there.
- The lab must be performed individually in class, but the report should be submitted in groups of up to 2 students.
- Lab groups must remain the same for all reports throughout the semester.

### Exercise 1:

We have a straight 8 × 8 P-box.
Table: 4 1 6 8 2 7 3 5. Input: 1 1 0 1 0 0 1 0.

1. Give the eight output bits.
2. Write the inverse table (swap contents and indices, then sort).
3. Apply the inverse table to your output and check you get the input back.

### Exercise 2 :

1. Let's have a 6 × 4 P-box that has table **5 3 2 6**. Apply it to **1 0 1 1 0 1**. Which input bits are lost? What do we call this P-Box?
2. A 4 × 6 P-box has table 4 1 2 3 4 1. Apply it to **1 0 1 1**. Which input bits are copied twice? What do we call this P-Box?
3. Can these two P-boxes be inverted. Why?

### Exercise 3:

A straight 6 × 6 P-box has table 2 5 3 6 1 4.

1. Find the inverse table.
2. A ciphertext block 0 1 0 0 1 1 came out of the P-box. Use the inverse table to find the input.

### Exercise 4

**A 3 × 2 S-box.** Row = leftmost bit, column = two rightmost bits.

| **Row \ Col** | **00** | **01** | **10** | **11** |
|---|---|---|---|---|
| 0 | 11 | 01 | 00 | 10 |
| 1 | 01 | 10 | 11 | 00 |

1. Encrypt 011, 110, 100 and 001.
2. Two of these inputs give the same output. What does that tell you about inverting this S-box?

### Exercise 5

**An invertible 3 × 3 S-box.** Encryption table:

| **Row \ Col** | **00** | **01** | **10** | **11** |
|---|---|---|---|---|
| 0 | 101 | 011 | 000 | 110 |
| 1 | 111 | 001 | 100 | 010 |

1. Encrypt 010 and 111.
2. Build the full decryption table (same layout: row = leftmost bit of the ciphertext).
3. Decrypt 100 and 011.

### Exercise 7

Choosing e. Bob picks p = 7 and q = 11.

1. Compute n and Φ(n).
2. Which of these are valid choices for e: 2, 7, 9, 11, 13, 15, 25? Justify with gcd.
3. For e = 7, find d.

### Exercise 8

p = 3, q = 11, e = 7.

1. Compute n, Φ(n) and d.
2. Give the public key and the private key.
3. Encrypt P = 2.
4. Decrypt your ciphertext and check you get 2 back.

### Exercise 9

Alice has p = 5, q = 11, e = 3, d = 27.

1. Which key does Alice use to sign, and which does Bob use to verify?
2. Sign the message M = 7.
3. Verify the signature as Bob would.

### Exercise 10

Ted sends the word HI to Jennifer, whose public key is (e = 7, n = 187). Letters are encoded A = 00 … Z = 25 and each letter is encrypted separately.

1. Encode H and I as numbers.
2. Encrypt each one.
3. Why can Ted not encode HI as the single number 0708 with this key?